

PANCREAS

Regular anatomy notes - Gallbladder-style layout with larger source diagrams

Definition and nature

- The pancreas is a soft, finely lobulated, elongated mixed gland located deeply in the upper abdomen.
- It is both exocrine and endocrine: the exocrine part secretes pancreatic juice, while the endocrine islets secrete hormones such as insulin and glucagon.
- Its juice helps in digestion of proteins, fats, and carbohydrates; endocrine secretions regulate glucose homeostasis.

High-yield orientation

- It lies behind the stomach and forms part of the posterior wall of the lesser sac.
- Most of the gland is retroperitoneal, but the tail lies in the lienorenal ligament.
- Because it is deeply placed and covered by other organs, the normal pancreas cannot be palpated during routine physical examination.

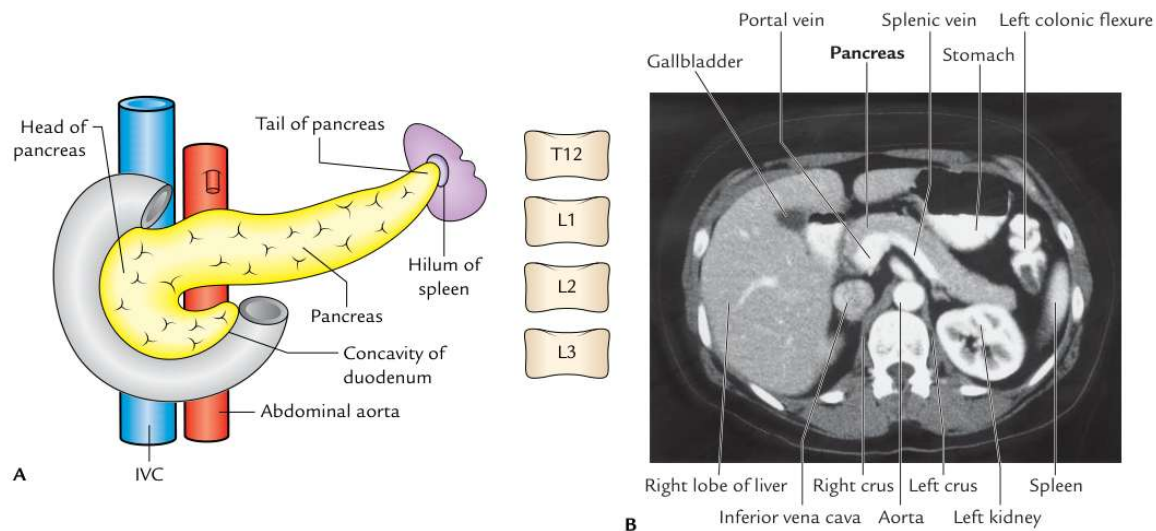


Fig. 9.12 Location of the pancreas: A, head lies opposite L1 and L3 vertebrae, body lies more or less in transpyloric plane (lower border of L1), and tail is at the level of T12 vertebra; B, CT scan of abdomen showing location of the pancreas. (Source: Fig. 4.88A, Page 289, *Gray's Anatomy for Students*, Richard L Drake, Wayne Vogl, Adam WM Mitchell. Copyright Elsevier Inc. 2005, All rights reserved.)

Source diagram: location of pancreas with CT correlation.

The pancreas crosses the posterior abdominal wall from the concavity of the duodenum to the hilum of the spleen. Its body lies approximately near the transpyloric plane, while the head and tail occupy different vertebral levels. The CT image helps correlate the pancreas with the stomach, portal vein, splenic vein, aorta, IVC, spleen, and kidneys.

1. Location, Size, Shape, and Parts

Location

The pancreas lies almost horizontally and slightly obliquely on the posterior abdominal wall in the epigastric and left hypochondriac regions. It extends from the C-shaped curve of the duodenum on the right to the hilum of the spleen on the left. Its greater part is retroperitoneal, lying behind the serous floor of the lesser sac.

Size and shape

- Shape: J-shaped or retort-shaped gland.
- Length: about 12-15 cm.
- Width: about 3-4 cm.
- Thickness: about 1.5-2 cm.
- Weight: about 80-90 g.

Parts

- Head, including the uncinate process.
- Neck: constricted portion connecting head and body.
- Body: elongated middle part with tuber omentale.
- Tail: narrow left extremity reaching the spleen.

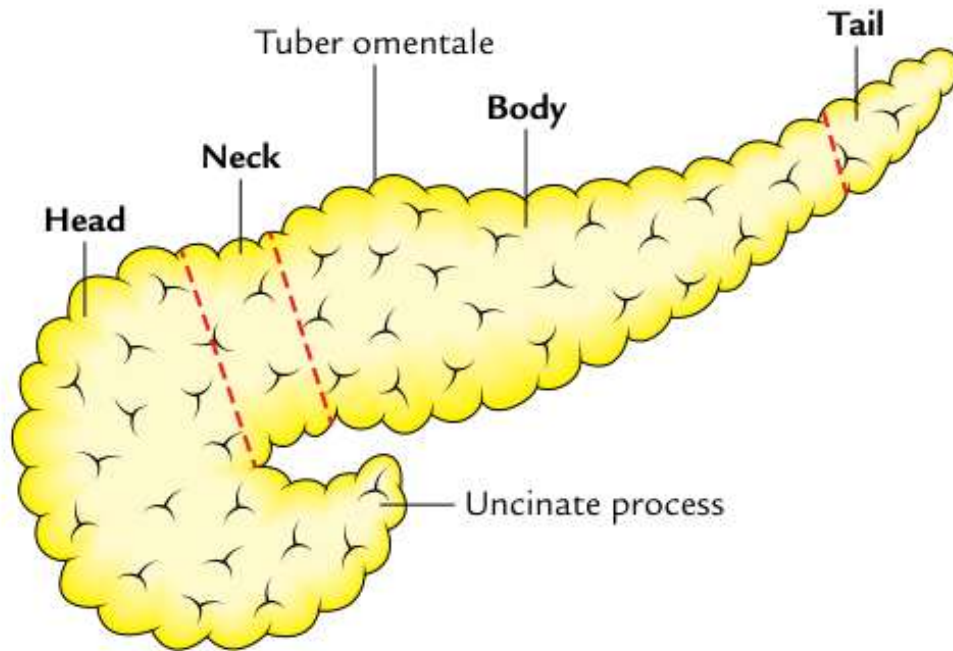


Fig. 9.13 Parts of the pancreas.

Source diagram: parts of the pancreas.

The head is broad and lies within the duodenal curve. The uncinate process projects from the lower part of the head behind the superior mesenteric vessels. The neck is short and constricted, the body forms the long central portion, and the tail is the narrow left end near the spleen.

Exam note on parts

When identifying the pancreas in a diagram, first locate the duodenal loop and the spleen. The head fits into the duodenal curve, the neck lies anterior to the formation of the portal vein, the body crosses the midline, and the tail reaches the splenic hilum within the lienorenal ligament.

2. Head of the Pancreas

General description

The head is the enlarged disc-shaped right end of the pancreas. It lies in the concavity of the C-shaped loop of the duodenum, in front of the L2 vertebra. The uncinate process is a hook-like projection from the lower and left part of the head and extends behind the superior mesenteric vessels.

External features of head

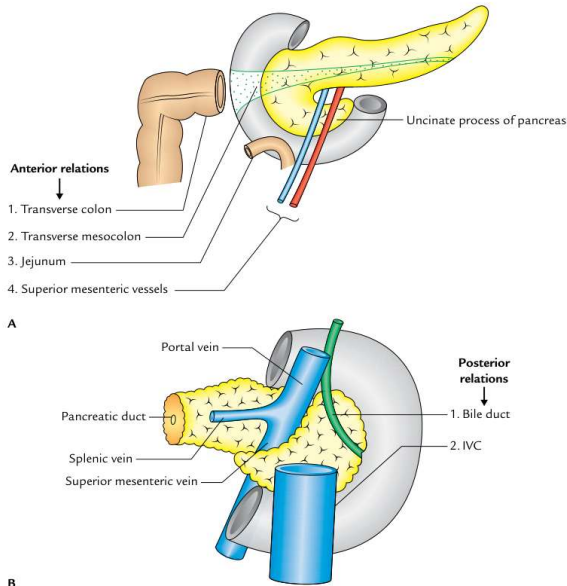
- Three borders: superior, inferior, and right lateral.
- Two surfaces: anterior and posterior.
- One process: uncinete process.

Relations of borders

- Superior border: first part of duodenum and superior pancreaticoduodenal artery.
- Inferior border: third part of duodenum and inferior pancreaticoduodenal artery.
- Right lateral border: second part of duodenum and pancreaticoduodenal arterial arcades.

Surfaces: Anterior and posterior.

(a) gastroduodenal artery,



Relations of the head of pancreas: A, anterior relations; B, posterior relations.

Source diagram: anterior and posterior relations of the head.

Diagram explanation

Anterior relations include gastroduodenal artery, transverse colon, root of transverse mesocolon, jejunum, and superior mesenteric vessels. Posterior relations include the bile duct and IVC. This close relation explains why carcinoma of the head may compress the bile duct and cause persistent obstructive jaundice.

Clinical importance of the head

The bile duct is closely related to the posterior aspect of the head and may even be embedded in pancreatic tissue. In pancreatitis or carcinoma of the head of pancreas, compression of the bile duct can produce obstructive jaundice. If the obstruction is malignant and gradual, the gallbladder may become distended and palpable.

3. Neck, Body, and Tail

Neck of pancreas

- Slightly constricted part connecting head with body.
- About 2.5 cm long and directed forwards, upwards, and to the left.
- Anterior surface is related to pylorus.
- Posterior surface is related to commencement of portal vein.
- Upper border relates to first part of duodenum; lower border to root of transverse mesocolon.

Body of pancreas

- Elongated part extending from neck to tail.
- Runs left of midline with slight upward and backward inclination.
- Lies in front of vertebral column at or just below the transpyloric plane.
- Triangular in cross-section with three borders and three surfaces.
- Tuber omentale projects above the lesser curvature of stomach.

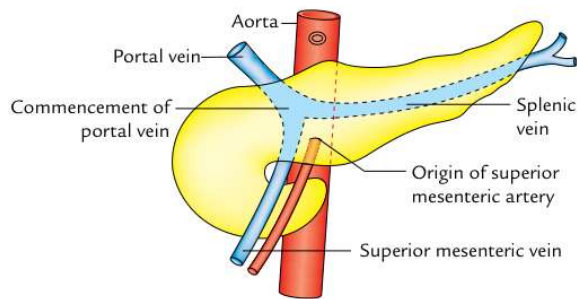


Fig. 9.15 Posterior relations of the neck of pancreas.

Source diagram: *posterior relations of the neck.*

Diagram explanation

The neck is a high-yield landmark because the portal vein begins behind it by union of the superior mesenteric vein and splenic vein. This explains why the neck is important in imaging, portal venous anatomy, and surgery of the pancreatic head and duodenum.

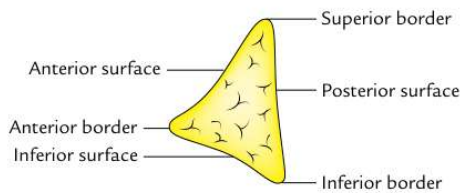


Fig. 9.16 Surfaces and borders of the body of pancreas as seen in cross section.

Source diagram: *cross-section of the body.*

Diagram explanation

The body is triangular in section. It has anterior, posterior, and inferior surfaces and anterior, superior, and inferior borders. Remember: the anterior border gives attachment to the transverse mesocolon, so the pancreas is closely tied to the lesser sac and transverse mesocolon region.

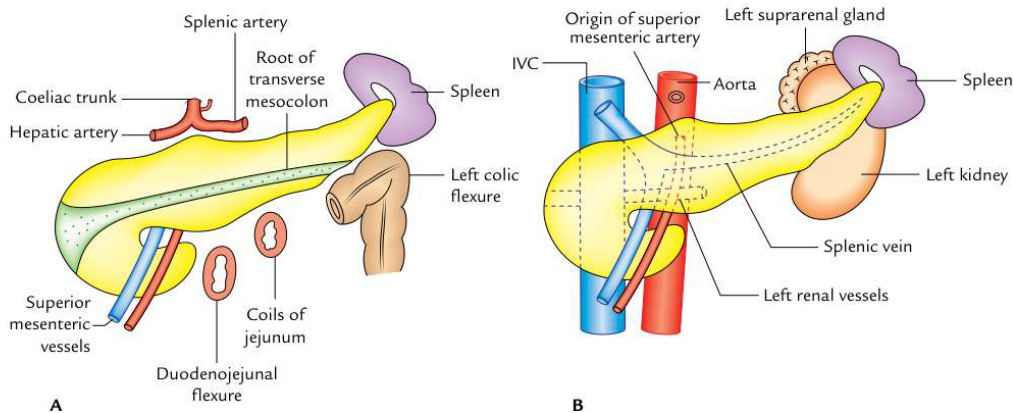


Fig. 9.17 Relations of the body of pancreas: A, anterior and inferior relations; B, posterior relations.

(b) coils of jejunum, and

4. It contains the largest number of islets of Langerhans

Source diagram: *relations of the body of pancreas.*

The anterior surface is related to the lesser sac and stomach. The posterior surface is non-peritoneal and related to the aorta, origin of the superior mesenteric artery, left kidney, left suprarenal gland, and splenic vein. The inferior surface is related to duodenojejunal flexure, jejunal coils, and left colic flexure.

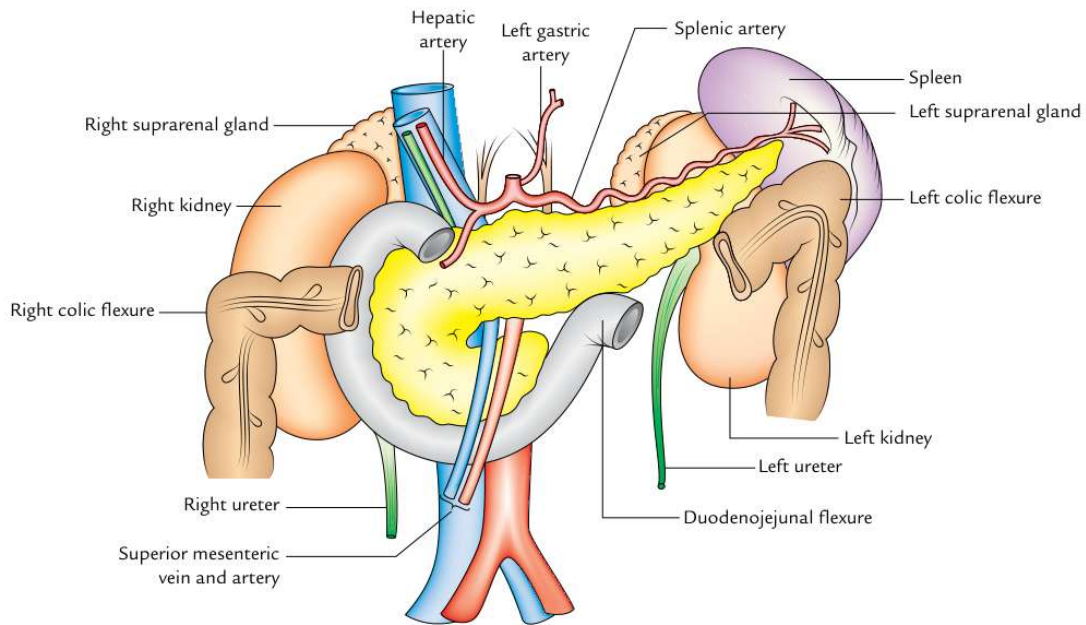


Fig. 9.18 Visceral relations of the different parts of the pancreas.

Source diagram: visceral relations of the different parts of pancreas.

This composite relation diagram is useful for long answers. The head is related to duodenum, bile duct, IVC, and superior mesenteric vessels. The body lies behind the stomach and lesser sac. The tail reaches the spleen and is the only mobile major part of the pancreas.

Tail of pancreas

The tail is the narrow left extremity of the gland. It lies in the lienorenal ligament along with the splenic vessels and reaches the visceral surface of the spleen between the gastric and colic impressions. It contains the largest number of islets of Langerhans per unit tissue, making it especially relevant to endocrine function.

4. Ducts of the Pancreas

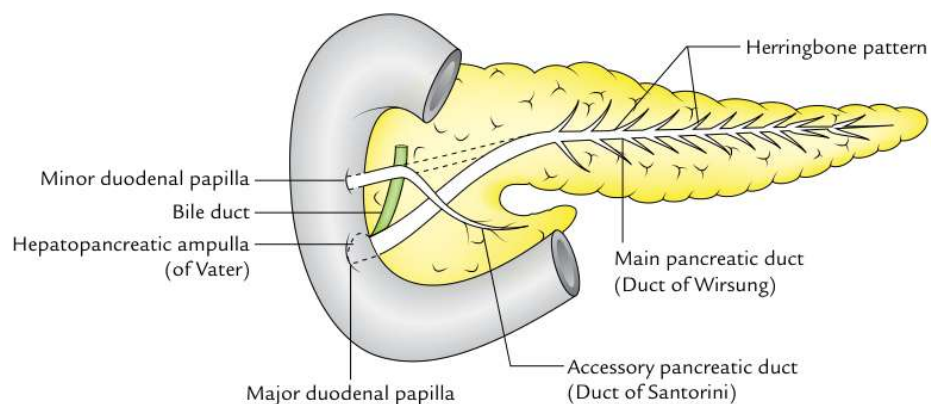


Fig. 9.19 Pancreatic ducts.

Source diagram: main and accessory pancreatic ducts.

The main pancreatic duct begins in the tail and passes through the gland near its posterior surface. It joins the bile duct to form the hepatopancreatic ampulla, opening at the major duodenal papilla. The accessory pancreatic duct may drain part of the head and opens at the minor duodenal papilla.

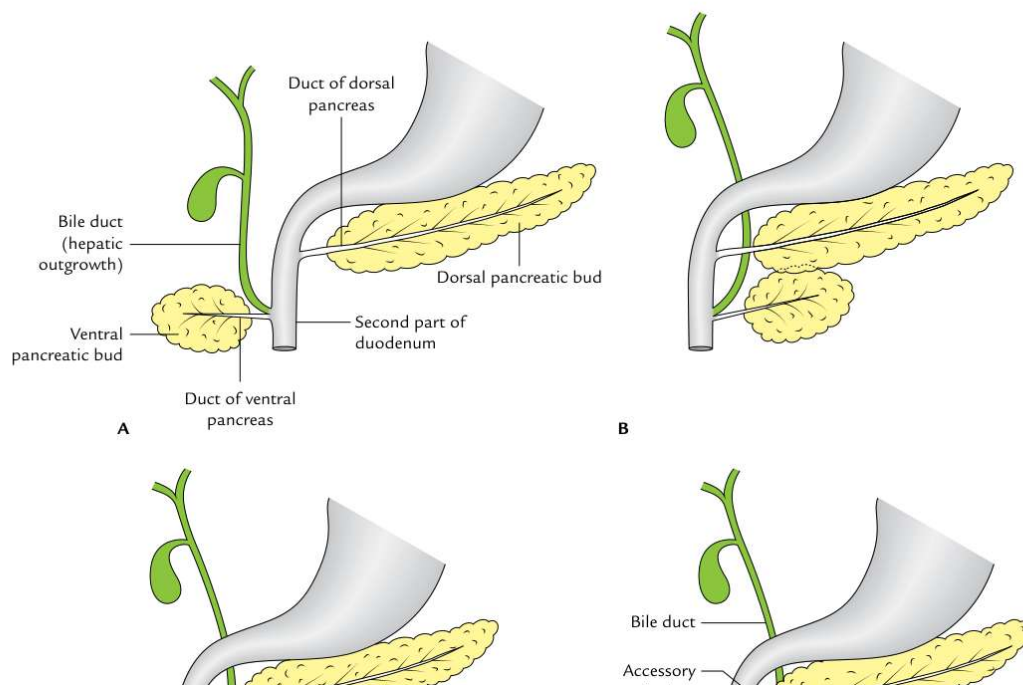
Ducts of Pancreas - Comparison

Feature	Main pancreatic duct	Accessory pancreatic duct
Other name	Duct of Wirsung	Duct of Santorini
Beginning	Begins in the tail and traverses the entire gland	Begins in the lower part of head
Course	Near posterior surface; turns down and right near the neck	Runs upward and medially, crossing in front of main duct
Opening	Major duodenal papilla through hepatopancreatic ampulla	Minor duodenal papilla above the major papilla
Clinical/exam note	Only duct in many cases; tributaries join in herringbone pattern	May communicate with main duct in about 40% of cases

High-yield duct point

The main pancreatic duct and bile duct unite near the second part of the duodenum. Therefore, obstruction near the ampulla can affect both bile drainage and pancreatic secretion. This is why the anatomy of the major duodenal papilla is important in jaundice, pancreatitis, and endoscopic procedures.

5. Development and Congenital Anomalies



Source diagram: development of pancreas from dorsal and ventral buds.

The pancreas develops from dorsal and ventral pancreatic buds. After gut rotation, the ventral bud moves posteriorly and fuses with the dorsal bud. The ventral bud contributes to the posterior part of head and uncinate process, while the dorsal bud forms the remaining gland. Ductal fusion forms the main and accessory ducts.

Development summary

- Ventral bud arises with the hepatic bud and bile duct.
- Dorsal bud arises more proximally from the duodenum.
- Ventral bud rotates behind the duodenum to fuse with the dorsal bud.
- Duct of ventral bud forms proximal part of main pancreatic duct.
- Duct of dorsal bud forms distal main duct and accessory duct.

Congenital anomalies

- Annular pancreas: ring of pancreatic tissue encircles the second part of duodenum.
- Accessory pancreatic tissue may occur in stomach, duodenum, small intestine, Meckel diverticulum, greater omentum, or splenic hilum.
- Accessory nodules are usually small yellowish lobulated nodules; some may contain islets.

Clinical relevance of annular pancreas

Annular pancreas can constrict the second part of the duodenum. In neonates or infants it may present with features of duodenal obstruction, vomiting, and feeding difficulty. In exams, always connect this anomaly with abnormal rotation/fusion of pancreatic buds.

6. Blood Supply, Venous Drainage, Lymphatics, and Nerves

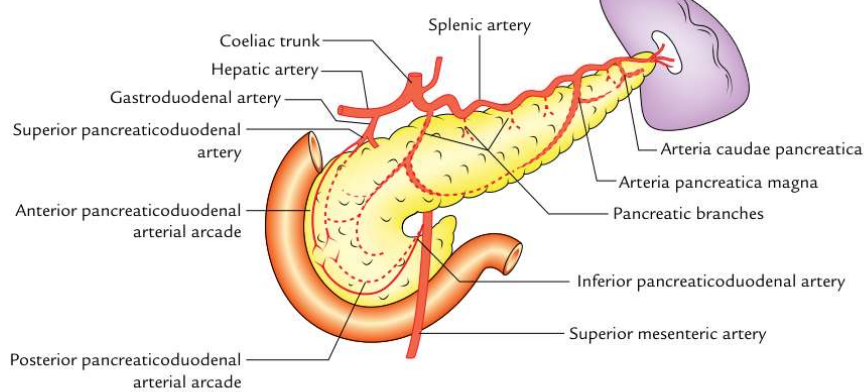


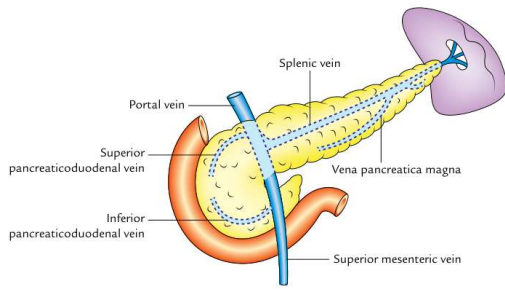
Fig. 9.21 Arterial supply of the pancreas.

Source diagram: arterial supply of the pancreas.

The pancreas is highly vascular. The splenic artery supplies body and tail through pancreatic branches including arteria pancreatica magna and arteria caudae pancreatica. The head receives superior and inferior pancreaticoduodenal arteries, which form anterior and posterior arterial arcades.

Arterial Supply

Region/part	Main arterial sources
Body and tail	Splenic artery branches; named branches include arteria pancreatica magna and arteria caudae pancreatica.
Head	Superior pancreaticoduodenal artery from gastroduodenal artery and inferior pancreaticoduodenal artery from superior mesenteric artery.
Arcades	Anterior and posterior pancreaticoduodenal arterial arcades form between the superior and inferior pancreaticoduodenal arteries.
Embryological reason	Pancreas develops near foregut-midgut junction, so it receives supply from coeliac trunk and superior mesenteric artery systems.



.22 Venous drainage of the pancreas.

Source diagram: venous drainage.

Diagram explanation

Veins of the pancreas drain into the portal vein, superior mesenteric vein, and splenic vein. This places pancreatic venous blood in the portal circulation and explains the importance of portal venous anatomy in pancreatic disease.

Lymphatic drainage

- Lymphatics follow arteries.
- Main group: pancreaticosplenic nodes.
- Other groups: coeliac nodes, superior mesenteric nodes, and pyloric nodes.
- Lymphatic spread is clinically important in pancreatic carcinoma.

Nerve supply

- Fibres reach the gland along arteries from coeliac and superior mesenteric plexuses.
- Sympathetic fibres are mainly vasomotor.
- Parasympathetic fibres control pancreatic secretion.
- Pain from the pancreas often radiates to the back because of its posterior position.

7. Applied Anatomy

Carcinoma of head of pancreas

- Common site: head of pancreas.
- It may compress the bile duct, causing persistent obstructive jaundice.
- It may press on the portal vein or involve the stomach because of close anatomical relations.
- Painless progressive jaundice with a distended gallbladder is a classic clinical clue when the obstruction is malignant.

Acute pancreatitis

- Acute inflammation may follow obstruction of the pancreatic duct, alcohol intake, viral infection such as mumps, or trauma.
- Activated pancreatic enzymes may leak into pancreatic tissue and cause autodigestion of the gland.
- Pain is severe in the epigastrium and commonly radiates to the back.
- Clinical features may include fever, nausea, vomiting, and raised serum amylase.

Why pancreatic pain radiates to the back

- The pancreas lies retroperitoneally on the posterior abdominal wall.
- Inflammation irritates posterior structures and retroperitoneal nerves.
- Therefore pain is classically epigastric with radiation to the back.

Why jaundice occurs

- The bile duct is closely related to the head of pancreas.
- Tumor or inflammation can compress the duct.
- Bile retention leads to obstructive jaundice and dark urine/pale stools clinically.

8. Quick Revision Table

Golden Points

Question	Answer
Nature of pancreas	Mixed exocrine-endocrine gland

Regions occupied	Epigastric and left hypochondriac regions
Shape	J-shaped / retort-shaped
Parts	Head, neck, body, tail; head has uncinete process; body has tuber omentale
Most retroperitoneal part	Most of the gland except the tail
Mobile part	Tail, because it lies in the lienorenal ligament
Relation behind neck	Commencement of portal vein
Main duct	Duct of Wirsung
Accessory duct	Duct of Santorini
Main blood supply of body/tail	Splenic artery branches
Important carcinoma site	Head of pancreas
Developmental anomaly	Annular pancreas

Long-answer writing plan

Start with definition, location, size and shape. Then describe parts one by one: head, neck, body, and tail. Add diagrams for parts, ducts, development, and blood supply. Finish with venous drainage, lymphatics, nerve supply, and applied anatomy such as carcinoma of head, obstructive jaundice, annular pancreas, and acute pancreatitis.

Diagram checklist

- Draw and label head, neck, body, tail, uncinete process, and tuber omentale.
- Show the pancreatic duct joining the bile duct at the major duodenal papilla.
- Draw arterial supply with splenic artery and pancreaticoduodenal arcades.

Common mistakes to avoid

- Do not forget that the tail is intraperitoneal/mobile in the lienorenal ligament.
- Do not confuse superior mesenteric vessels with splenic vessels.
- Do not miss the portal vein forming behind the neck.
- Do not write only exocrine function; endocrine function is equally important.

Integrated clinical summary for last-page revision

The pancreas is clinically important because of three close relations: the head is related to the bile duct, the neck is related posteriorly to the portal vein formation, and the body lies across major vessels on the posterior abdominal wall. Disease of the head commonly presents with obstructive jaundice, while pancreatitis classically causes severe epigastric pain radiating to the back. In practical anatomy, always connect the duct system with the second part of the duodenum and the blood supply with both coeliac and superior mesenteric systems.

Five lines to remember

- Head lies in the duodenal curve.
- Uncinate process passes behind superior mesenteric vessels.
- Portal vein begins behind the neck.
- Tail lies in lienorenal ligament and reaches spleen.
- Main pancreatic duct joins bile duct at the hepatopancreatic ampulla.

Applied anatomy triggers

- Head tumor: obstructive jaundice.
- Duct obstruction: pancreatitis.
- Tail injury: splenic vessel risk.
- Annular pancreas: duodenal obstruction.
- Portal vein relation: surgical and imaging importance.

Source base: Vishram Singh, Textbook of Anatomy: Abdomen and Lower Limb, Volume II. Diagrams are adapted from the uploaded source PDF for study use.

